

Graphify Report — Cargo Delivery Service

Generated: 2026-06-14
Repository: D: **Analysis Scope:** Full codebase (architecture docs, frontend source, mock data, status mapping engine)

1. Executive Summary

Centralized multi-carrier logistics platform unifying **DHL, FedEx, UPS, Chronopost** under a single `CarrierAdapter` interface. The core engine is a 3-tier status mapper (exact → regex → fuzzy) translating 64+ carrier codes into 12 unified internal codes. The React + Vite + Tailwind TypeScript SPA provides full CRUD, JWT mock authentication, and trilingual support (EN/FR/TR).

2. API — Complete Endpoints with JSON

2.1 Reference Table (20 endpoints)

#	Method	Path	Purpose	F
1	POST	/api/v1/cargo/auth/login	Authenticate & get JWT token	Log
2	POST	/api/v1/cargo/rates	Get carrier pricing	Rate
3	POST	/api/v1/cargo/shipments	Create a shipment	Ship
4	GET	/api/v1/cargo/shipments	List shipments	Ship
5	GET	/api/v1/cargo/shipments/{id}	Shipment detail + status history	Ship
6	GET	/api/v1/cargo/shipments/{id}/tracking	Detailed tracking events	Ship
7	POST	/api/v1/cargo/shipments/{id}/cancel	Cancel a shipment	Ship

#	Method	Path	Purpose	F
8	POST	/api/v1/cargo/shipments/{id}/label	Generate shipping label	Ship
9	POST	/api/v1/cargo/pickups	Schedule a pickup	Pick
10	POST	/api/v1/cargo/webhooks/{carrierCode}	Receive carrier push updates	Web
11	POST	/api/v1/cargo/addresses/validate	Validate & normalize address	Ship form
12	POST	/api/v1/cargo/carriers	Add a carrier	Carr
13	GET	/api/v1/cargo/carriers	List carriers	Carr
14	GET	/api/v1/cargo/carriers/{code}	Carrier detail	Carr
15	PUT	/api/v1/cargo/carriers/{code}	Update carrier settings	Carr
16	PATCH	/api/v1/cargo/carriers/{code}/toggle	Enable/disable carrier	Carr
17	POST	/api/v1/cargo/carriers/{code}/test	Test carrier API connection	Carr
18	POST	/api/v1/cargo/carriers/{code}/services	Add a service	Carr
19	PUT	/api/v1/cargo/carriers/{code}/credentials	Update API keys	Carr
20	DELETE	/api/v1/cargo/carriers/{code}	Delete a carrier	Carr

2.2 POST — Creation / Action

POST /api/v1/cargo/auth/login

Authenticate user and return a mock JWT token.

```
// Request
{ "email": "admin@cargo.com", "password": "admin" }

// Response 200
{
  "success": true,
  "user": { "email": "admin@cargo.com", "name": "admin", "role": "admin" },
  "token": "mock-jwt-a1b2c3d4"
}
```

POST /api/v1/cargo/rates

Compare prices from all active carriers for a shipment.

// Request

```
{
  "sender": { "country": "FR", "zipCode": "75001", "city": "Paris",
    "address": "12 Rue de Rivoli" },
  "recipient": { "country": "DE", "zipCode": "10115", "city": "Berlin",
    "address": "Friedrichstraße 100" },
  "packages": [{ "weight": 2.5, "weightUnit": "KG", "length": 30, "width":
    20, "height": 15, "dimUnit": "CM" }],
  "options": { "carrierCodes": ["DHL", "FEDEX"], "serviceType": "EXPRESS" }
}
```

// Response 200

```
{
  "success": true,
  "data": [
    {
      "carrierCode": "DHL", "carrierName": "DHL Express",
      "serviceCode": "DHL_EXPRESS_WORLDWIDE", "serviceName": "Express
        Worldwide",
      "totalPrice": 45.30, "currency": "EUR",
      "estimatedTransitDays": 2, "estimatedDeliveryDate": "2026-06-13",
      "breakdown": [
        { "type": "BASE", "label": "Transport", "amount": 38.00 },
        { "type": "FUEL", "label": "Fuel surcharge", "amount": 5.30 },
        { "type": "INSURANCE", "label": "Insurance", "amount": 2.00 }
      ]
    },
    {
      "carrierCode": "FEDEX", "carrierName": "FedEx",
      "serviceCode": "FEDEX_PRIORITY", "serviceName": "FedEx Priority",
      "totalPrice": 48.50, "currency": "EUR",
      "estimatedTransitDays": 1, "estimatedDeliveryDate": "2026-06-12",
      "breakdown": [
        { "type": "BASE", "label": "Transport", "amount": 41.00 },
        { "type": "FUEL", "label": "Fuel surcharge", "amount": 5.50 },
        { "type": "INSURANCE", "label": "Insurance", "amount": 2.00 }
      ]
    }
  ]
}
```

POST /api/v1/cargo/shipments

Create a real shipment with a carrier. Payload: sender + recipient + packages + options.

// Request (20+ fields)

```
{
```

```

"carrierCode": "DHL", "serviceCode": "DHL_EXPRESS_WORLDWIDE", "reference":
  "ORD-2026-001234",
"sender": { "company": "TechCorp Inc", "contactName": "John Doe", "email":
  "john@techcorp.com",
  "phone": "+33123456789", "country": "FR", "zipCode": "75001", "city":
  "Paris", "address": "12 Rue de Rivoli" },
"recipient": { "company": "Berlin GmbH", "contactName": "Hans Schmidt",
  "email": "hans@berlin.de",
  "phone": "+4930123456", "country": "DE", "zipCode": "10115", "city":
  "Berlin", "address": "Friedrichstraße 100" },
"packages": [{
  "reference": "PARCEL-001", "description": "Electronic components",
  "weight": 2.5, "weightUnit": "KG", "length": 30, "width": 20, "height":
  15, "dimUnit": "CM",
  "declaredValue": 500.00, "declaredCurrency": "EUR"
}],
"options": { "insurance": { "amount": 500.00, "currency": "EUR" },
  "signatureRequired": true }
}

// Response 201
{
  "success": true,
  "data": {
    "id": "shp_a1b2c3d4e5", "status": "CREATED",
    "carrierCode": "DHL", "carrierTrackingNumber": "1234567890",
    "labelUrl": "https://api.dhl.com/labels/1234567890.pdf",
    "trackingUrl": "https://www.dhl.com/track/1234567890",
    "price": { "total": 45.30, "currency": "EUR", "breakdown": [
      { "type": "BASE", "amount": 38.00 }, { "type": "FUEL", "amount": 5.30
      }, { "type": "INSURANCE", "amount": 2.00 }
    ]},
    "estimatedDeliveryDate": "2026-06-13T18:00:00Z",
    "createdAt": "2026-06-11T10:30:00Z"
  }
}

```

POST /api/v1/cargo/shipments/{id}/cancel

Cancel a shipment. Status changes to CANCELLED.

```

// Response
{
  "success": true,
  "data": { "id": "shp_a1b2c3d4e5", "status": "CANCELLED", "cancelledAt":
    "2026-06-11T12:00:00Z" }
}

```

POST /api/v1/cargo/shipments/{id}/label

Generate the shipping label in the requested format (PDF, ZPL).

```
// Request
{ "format": "PDF" }

// Response
{
  "success": true,
  "data": { "labelUrl": "https://api.dhl.com/labels/1234567890.pdf",
            "format": "PDF", "size": "A6", "generatedAt": "2026-06-11T10:30:00Z" }
}
```

POST /api/v1/cargo/pickups

Schedule a carrier pickup.

```
// Request
{
  "carrierCode": "DHL", "shipmentIds": ["shp_a1b2c3d4e5"],
  "pickupDate": "2026-06-12", "readyTime": "09:00", "closeTime": "17:00",
  "location": { "company": "TechCorp Inc", "contactName": "John Doe",
                "phone": "+33123456789",
                "country": "FR", "zipCode": "75001", "city": "Paris", "address": "12 Rue de Rivoli" },
  "totalPackages": 1, "totalWeight": 2.5, "weightUnit": "KG",
  "specialInstructions": "Ring at reception"
}

// Response
{
  "success": true,
  "data": {
    "id": "pck_001a2b3c", "carrierCode": "DHL", "carrierPickupId": "DHL-PICKUP-98765",
    "status": "CONFIRMED", "pickupDate": "2026-06-12",
    "readyTime": "09:00", "closeTime": "17:00",
    "confirmationNumber": "CONF-ABC-123"
  }
}
```

POST /api/v1/cargo/webhooks/{carrierCode}

Entry point for carrier push notifications. Always returns 200.

```
// Request (sent by DHL)
{
  "shipmentId": "1234567890", "status": "DELIVERED", "statusDescription": "Shipment delivered",
  "timestamp": "2026-06-13T16:30:00Z",
  "location": { "city": "Berlin", "country": "DE", "zipCode": "10115" },
  "signedBy": "Hans Schmidt", "deliveryPhotoUrl": "https://dhl.com/proof/photo.jpg"
}
```

```
// Response (always 200)
{ "success": true, "message": "Webhook processed successfully" }
```

POST /api/v1/cargo/addresses/validate

Validate and normalize an address via the carrier (real-time check).

```
// Request
{ "country": "FR", "zipCode": "75001", "city": "Paris", "address": "12 Rue de
  Rivoli", "carrierCode": "DHL" }
```

```
// Response
{
  "success": true,
  "data": {
    "valid": true,
    "normalizedAddress": { "country": "FR", "zipCode": "75001", "city":
      "PARIS", "address": "12 RUE DE RIVOLI" },
    "suggestions": [],
    "carrierValidation": { "code": "VALID", "message": "Address is valid" }
  }
}
```

POST /api/v1/cargo/carriers

Add a carrier with credentials, services, and capabilities.

```
// Request (simplified payload – see service_cargo.md for full)
{
  "code": "CHRONOPOST", "name": "Chronopost", "adapterName":
    "ChronopostAdapter",
  "active": true, "website": "https://www.chronopost.fr",
  "services": [
    { "code": "CHRONO_13", "name": "Chrono 13h", "zones": ["FR", "MC"],
      "transitDays": 1,
      "features": ["SIGNATURE", "INSURANCE"], "active": true }
  ],
  "credentials": {
    "authType": "API_KEY", "apiKey": "chrono_api_key_12345",
    "accountNumber": "CHRONO-ACC-001", "endpoint":
      "https://api.chronopost.fr/v1"
  },
  "settings": { "timeoutMs": 15000, "retryMaxAttempts": 3,
    "rateLimitPerMinute": 60 }
}
```

```
// Response 201
{
  "success": true,
  "data": {
    "code": "CHRONOPOST", "name": "Chronopost", "active": true,

```

```

    "status": "PENDING_TEST", "message": "Carrier added. Run a connection
    test before using."
  }
}

```

POST /api/v1/cargo/carriers/{code}/test

Test the carrier API connection.

// Response – Success

```

{
  "success": true,
  "data": {
    "status": "CONNECTED", "latencyMs": 245, "testedAt": "2026-06-
    11T15:00:00Z",
    "endpoint": "https://api.chronopost.fr/v1/health",
    "details": { "httpStatus": 200, "message": "API is reachable",
    "accountValid": true }
  }
}

```

// Response – Failure

```

{
  "success": false,
  "error": { "code": "CARRIER_CONNECTION_FAILED", "message": "Failed to
    connect to Chronopost API",
    "details": { "httpStatus": 401, "reason": "Invalid API key" } }
}

```

POST /api/v1/cargo/carriers/{code}/services

Add a service to the carrier's catalog.

// Request

```

{
  "code": "DHL_EXPRESS_9", "name": "Express 9:00",
  "description": "Delivery before 9am next day",
  "maxWeight": 30, "maxWeightUnit": "KG",
  "zones": ["FR", "DE", "BE", "NL", "LU"],
  "transitDays": 1, "features": ["SIGNATURE", "INSURANCE"], "active": true
}

```

2.3 GET — Read / Query

GET /api/v1/cargo/shipments

Paginated list of shipments. Filters: ?status=IN_TRANSIT&page=1&limit=20.

// Response

```

{

```

```
"success": true,
"data": [
  {
    "id": "shp_a1b2c3d4e5", "status": "IN_TRANSIT",
    "carrierCode": "DHL", "carrierTrackingNumber": "1234567890",
    "reference": "ORD-2026-001234",
    "sender": { "company": "TechCorp Inc", "country": "FR" },
    "recipient": { "company": "Berlin GmbH", "country": "DE" },
    "price": { "total": 45.30, "currency": "EUR" },
    "createdAt": "2026-06-11T10:30:00Z"
  }
],
"pagination": { "page": 1, "limit": 20, "total": 5 }
}
```

GET /api/v1/cargo/shipments/{id}

Full shipment detail with status history.

```
// Response
{
  "success": true,
  "data": {
    "id": "shp_a1b2c3d4e5", "status": "IN_TRANSIT",
    "statusHistory": [
      { "status": "CREATED", "timestamp": "2026-06-11T10:30:00Z" },
      { "status": "PICKED_UP", "timestamp": "2026-06-11T14:00:00Z" },
      { "status": "IN_TRANSIT", "timestamp": "2026-06-12T08:15:00Z" }
    ],
    "carrierCode": "DHL", "carrierTrackingNumber": "1234567890",
    "carrierShipmentId": "DE-2026-98765",
    "reference": "ORD-2026-001234",
    "labelUrl": "https://api.dhl.com/labels/1234567890.pdf",
    "trackingUrl": "https://www.dhl.com/track/1234567890",
    "sender": { "company": "TechCorp Inc", "country": "FR", "zipCode": "75001" },
    "recipient": { "company": "Berlin GmbH", "country": "DE", "zipCode": "10115" },
    "packages": [{ "reference": "PARCEL-001", "weight": 2.5, "trackingNumber": "1234567890-001" }],
    "price": { "total": 45.30, "currency": "EUR" },
    "createdAt": "2026-06-11T10:30:00Z",
    "estimatedDeliveryDate": "2026-06-13T18:00:00Z"
  }
}
```

GET /api/v1/cargo/shipments/{id}/tracking

Tracking events with location, raw carrier status, and milestones.

```
// Response
{
```



```

"success": true,
"data": {
  "shipmentId": "shp_a1b2c3d4e5", "carrierCode": "DHL",
  "carrierTrackingNumber": "1234567890",
  "currentStatus": {
    "code": "IN_TRANSIT", "label": "In transit",
    "location": "Frankfurt, Germany", "timestamp": "2026-06-12T08:15:00Z"
  },
  "estimatedDeliveryDate": "2026-06-13T18:00:00Z",
  "events": [
    { "eventId": "evt_001", "code": "LABEL_CREATED", "label": "Label
      created",
      "description": "Shipping label generated",
      "location": "Paris, France", "timestamp": "2026-06-11T10:30:00Z",
      "carrierRawStatus": "Shipment information received" },
    { "eventId": "evt_002", "code": "PICKED_UP", "label": "Parcel
      collected",
      "location": "Paris, France", "timestamp": "2026-06-11T14:00:00Z",
      "carrierRawStatus": "Pickup scanned" },
    { "eventId": "evt_003", "code": "IN_TRANSIT", "label": "In transit",
      "location": "Frankfurt, Germany", "timestamp": "2026-06-
        12T08:15:00Z",
      "carrierRawStatus": "Departed from transit hub" }
  ],
  "milestones": {
    "created": "2026-06-11T10:30:00Z", "pickedUp": "2026-06-11T14:00:00Z",
    "inTransit": "2026-06-12T08:15:00Z",
    "outForDelivery": null, "delivered": null
  }
}
}

```

GET /api/v1/cargo/carriers

Paginated carrier list. Filters: ?

active=true&zone=FR&feature=INSURANCE&page=1&limit=20.

// Response

```

{
  "success": true,
  "data": [
    {
      "code": "DHL", "name": "DHL Express", "adapterName": "DhlAdapter",
      "active": true, "status": "CONNECTED", "lastTestedAt": "2026-06-
        10T08:00:00Z",
      "services": [
        { "code": "DHL_EXPRESS_WORLDWIDE", "name": "Express Worldwide",
          "active": true },
        { "code": "DHL_EXPRESS_12", "name": "Express 12:00", "active": true }
      ],
      "capabilities": { "labelFormats": ["PDF", "ZPL", "PNG"], "features":
        ["RATES", "TRACKING", "SIGNATURE", "INSURANCE", "SATURDAY"] }
    }
  ]
}

```

```

    }
  ],
  "pagination": { "page": 1, "limit": 20, "total": 3 }
}

```

GET /api/v1/cargo/carriers/{code}

Full carrier detail.

// Response

```

{
  "success": true,
  "data": {
    "code": "DHL", "name": "DHL Express", "adapterName": "DhlAdapter",
    "active": true, "status": "CONNECTED", "lastTestedAt": "2026-06-
      10T08:00:00Z",
    "services": [{
      "code": "DHL_EXPRESS_WORLDWIDE", "name": "Express Worldwide",
      "description": "International express delivery",
      "maxWeight": 70, "zones": ["WORLDWIDE"], "transitDays": 2,
      "features": ["SIGNATURE", "INSURANCE", "SATURDAY"], "active": true
    }],
    "capabilities": {
      "labelFormats": ["PDF", "ZPL", "PNG"],
      "features": ["RATES", "TRACKING", "SIGNATURE", "INSURANCE", "SATURDAY",
        "PICKUP"]
    },
    "settings": { "timeoutMs": 10000, "retryMaxAttempts": 3,
      "rateLimitPerMinute": 100, "webhookEnabled": true }
  }
}

```

2.4 PUT — Update

PUT /api/v1/cargo/carriers/{code}

Update carrier settings.

// Request

```

{ "settings": { "timeoutMs": 15000, "retryMaxAttempts": 3 }, "active": true }

```

PUT /api/v1/cargo/carriers/{code}/credentials

Update API keys and secrets. Re-test connection after update.

// Request

```

{
  "authType": "API_KEY", "apiKey": "new_dhl_api_key_67890",
  "apiSecret": "new_dhl_secret_12345", "webhookSecret": "whsec_new_abc456"
}

```

```
// Response
{
  "success": true,
  "data": {
    "code": "DHL", "credentialsUpdatedAt": "2026-06-11T16:00:00Z",
    "message": "Credentials updated. Re-test connection."
  }
}
```

2.5 PATCH — Partial Update

PATCH /api/v1/cargo/carriers/{code}/toggle

Enable or disable a carrier. Useful during outages or maintenance.

```
// Request
{ "active": false, "reason": "API outage - maintenance" }

// Response
{
  "success": true,
  "data": { "code": "FEDEX", "active": false, "deactivatedAt": "2026-06-11T14:00:00Z" }
}
```

2.6 DELETE — Deletion

DELETE /api/v1/cargo/shipments/{id}

Permanently delete a shipment.

```
// Response
{ "success": true, "data": { "deleted": true } }
```

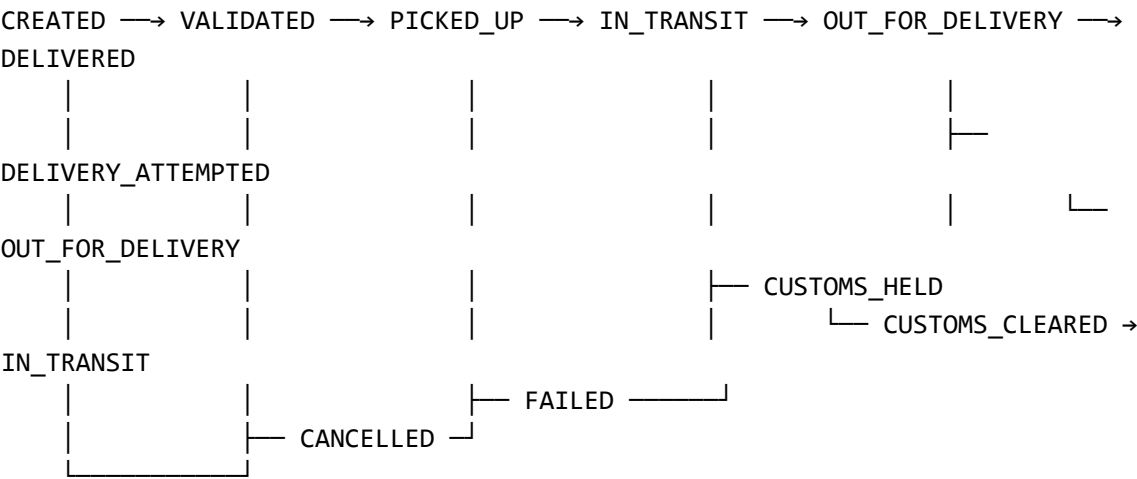
DELETE /api/v1/cargo/carriers/{code}

Delete a carrier.

```
// Response
{ "success": true, "data": { "deleted": true } }
```

3. Workflow

3.1 Shipment Lifecycle



7 unified statuses: CREATED → VALIDATED → PICKED_UP → IN_TRANSIT → OUT_FOR_DELIVERY → DELIVERED (terminal: DELIVERED, FAILED, CANCELLED)

3.2 Complete Workflow (10 steps)

- | | |
|--|---------------------------|
| 1. POST /api/v1/cargo/carriers | → Add Chronopost |
| 2. POST /api/v1/cargo/carriers/CHRONOPOST/test | → Test connection |
| 3. GET /api/v1/cargo/carriers | → List available carriers |
| 4. POST /api/v1/cargo/rates | → Compare DHL / FedEx |
| prices | |
| 5. POST /api/v1/cargo/addresses/validate | → Validate recipient |
| address | |
| 6. POST /api/v1/cargo/shipments | → Create shipment |
| 7. POST /api/v1/cargo/pickups | → Schedule pickup |
| 8. ← DHL Webhook (DELIVERED) | → Automatic status update |
| 9. GET /api/v1/cargo/shipments/{id}/tracking | → Track in real time |
| 10. POST /api/v1/cargo/shipments/{id}/cancel | → Cancel if needed |

3.3 Frontend ↔ Endpoint Mapping

- | | |
|-----------------------|--|
| Login (Page) | → POST auth/login |
| Rates (Page) | → POST cargo/rates |
| Shipments (Page) | → GET cargo/shipments POST cargo/shipments DELETE cargo/shipments/{id} |
| ShipmentDetail (Page) | → GET cargo/shipments/{id} POST .../cancel POST .../label GET .../tracking |
| Carriers (Page) | → GET cargo/carriers POST cargo/carriers DELETE cargo/carriers/{code} |
| CarrierDetail (Page) | → GET cargo/carriers/{code} PUT cargo/carriers/{code} |
| | PATCH .../toggle POST .../test POST |

```

.../services | PUT .../credentials
Pickups (Page) → POST cargo/pickups | POST
cargo/pickups/{id}/cancel
WebhookLogs (Page) → POST cargo/webhooks/{carrierCode} (logs)
Dashboard (Page) → Aggregates stats from shipments + carriers

```

3.4 Data Models

Table shipments (20 fields): id, status (8-value enum), carrier_code, carrier_tracking_number, reference, label_url, sender (JSON), recipient (JSON), packages (JSON[]), price (decimal), insurance_amount, estimated_delivery_date, dates...

Table tracking_events (11 fields): id, shipment_id, code (12-value enum: LABEL_CREATED, PICKED_UP, IN_TRANSIT, ARRIVED_AT_HUB, DEPARTED_FROM_HUB, OUT_FOR_DELIVERY, DELIVERED, DELIVERY_ATTEMPTED, FAILED, RETURNED_TO_SENDER, CUSTOMS_HELD, CUSTOMS_CLEARED), location (JSON), carrier_raw_status, signed_by, delivery_photo_url...

Table carriers (12 fields): code (PK), name, adapter_name, active, status (enum: PENDING_TEST, CONNECTED, DISCONNECTED, ERROR), capabilities (JSON), settings (JSON), credentials (encrypted JSON)...

Table carrier_services (9 fields): id, carrier_code (FK), code, name, max_weight, zones (JSON[]), transit_days, features (JSON[]), active

4. Knowledge Graph Analysis

4.1 Statistics

Metric	Value
Total Nodes	40
Total Edges	48
Communities	4
EXTRACTED edges	43 (89.6%)
INFERRED edges	5 (10.4%)
Average confidence score	0.949

4.2 Community Map

Community 0 — Architecture & Adapters (7 nodes, blue)

CarrierAdapter interface implemented by 4 adapters (DHL, FedEx, UPS, Chronopost). Each adapter normalizes carrier concepts into UnifiedStatusModel.

Community 1 — Status Mapping Engine (6 nodes, green)

StatusMapperEngine with 3-level per-carrier matching. 64+ raw codes → 12 unified codes.

Carrier	Codes	Exact	Regex	Fuzzy
DHL	15	8	5	2
FedEx	16	9	5	2
UPS	18	10	6	2
Chronopost	15	7	5	3

Community 2 — Frontend (21 nodes, yellow)

React 18 + TypeScript + Vite + Tailwind. Layered architecture: Pages → Components → Hooks → Service Layer → Mock Data.

Community 3 — API Design & Documentation (7 nodes, purple)

20 endpoints covering 6 domains: shipments, carriers, rates, pickups, webhooks, address validation.

4.3 God Nodes (High-Connectivity Hubs)

Node	Community	Degree	Type
cargo_service_layer	Frontend	9	code
status_mapper	Mapping	6	code
react_frontend	Frontend	6	code
rest_endpoints	API Design	6	document
cargo_service_arch	Architecture	5	document
unified_status	Architecture	5	code
shared_components	Frontend	5	code

cargo_service_layer is the most critical node — every page depends on it.

4.4 Edge Analysis

EXTRACTED (confidence 1.0): depends_on, contains, implements, uses, part_of, maps_to

INFERRED (confidence < 1.0): complements (status_mapper ↔ carrier_adapter, 0.85), crosscuts (il8n → frontend, 0.95), links_to (dashboard → shipments, 0.95), competes_with (DHL ↔ FedEx, 0.85)

4.5 Architectural Insights

- 1. **Modular architecture** — Adding a 5th carrier = implement interface + add status maps
- 2. **Status Mapping = key differentiator** — Turkish carrier status names supported via regex
- 3. **Service Layer = bottleneck** — Consider splitting into domain services (shipmentService, carrierService...)
- 4. **Frontend 52.5% of codebase** — Ready for real API by swapping cargoService.ts
- 5. **3 documentation files** — Drift risk. Auto-generation recommended

5. Project Files

Core Documents

File	Purpose
CARGO-SERVICE-ARCHITECTURE.md	Full backend architecture plan
service_cargo.md	API documentation (FR)
cargo_service.md	API documentation (EN)
cargo_service_api.md	Color-coded API reference

Frontend Source

File/Directory	Purpose
frontend/src/types/index.ts	14 TypeScript interfaces
frontend/src/data/mockData.ts	6 mock stores
frontend/src/services/statusMapper.ts	Status mapping engine (4 carriers)
frontend/src/services/cargoService.ts	25+ mock API functions
frontend/src/hooks/useCargoService.ts	useApi, useMutation, useLoading
frontend/src/context/AuthContext.tsx	JWT mock auth
frontend/src/i18n/	EN/FR/TR (750+ keys)
frontend/src/pages/	9 pages
frontend/src/components/	10+ shared components

Report generated by Graphify — knowledge graph pipeline.